

# FOL: A Functional Object Lisp

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## Abstract

We present FOL (Functional Object Lisp), a Lisp dialect combining Clojure’s persistent data structures with CLOS-style object orientation. Using this combination, we find that persistent objects and CLOS are not antagonistic but, in fact synergistic. Among other software capabilities, the combination yields metaobject-protocol-enabled versioned objects, declarative event sourcing using method combinations, and lazy schema evaluation—patterns more natural than in either parent alone. Benchmarks show that persistence adds only 1–3x overhead for sequential modifications. A preliminary transpiler to Common Lisp with minor optimizations reaches parity with hand-written Common Lisp code. FOL is written in Common Lisp atop FSet and Sycamore persistent data structure libraries, providing Clojure-compatible persistent collection objects and a meta-object-protocol adapted for immutable storage.

## CCS Concepts

• **Software and its engineering** → **Functional languages; Object oriented languages; Extensible languages; Language features.**

## Keywords

Functional programming, object-oriented programming, Lisp, persistent data structures, CLOS, MOP, transducers

## 1 Introduction

FOL explores the synergy between Common Lisp-style object orientation (CLOS) and Clojure-style immutable data structures. While CLOS offers powerful multiple dispatch, method combinations, and a metaobject protocol (MOP), it is fundamentally rooted in mutable state. Conversely, Clojure’s immutable model provides  $O(\log n)$  updates and structural sharing but lacks the introspective extensibility of CLOS. FOL merges these paradigms, demonstrating that persistent objects and CLOS are not only compatible but synergistic.

Immutable objects simplify multi-threaded development by eliminating synchronization conflicts and manual locking. FOL implements these objects by adapting the CLOS MOP to use immutable data structures for slot storage. We implemented FOL as a standalone Lisp rather than a library to ensure language-level correctness; a library-based approach would struggle to strictly enforce persistence invariants and manage the integration of persistent and primitive types.

The combination of immutability and CLOS enables patterns more natural than in either parent alone:

- **Versioned Objects via the MOP:** Functional updates transparently produce instances of the new schema while preserving old snapshots, trading CLOS’s explicit migration hooks for corruption safety.

- **Event Sourcing via Method Combinations:** Declarative :around methods intercept all mutations to build immutable audit logs automatically, a guarantee harder to enforce in Clojure.

Beyond architectural synergy, FOL provides concise predicate dispatch and enhanced destructuring. Our benchmarks show that FOL achieves a 15–28% reduction in LOC while maintaining performance parity with hand-written Common Lisp for complex workloads.

## 2 Language Design

### 2.1 Core Philosophy

FOL makes several design commitments:

- (1) **Immutability by default:** All objects and collections (except streams, atoms, etc.) are persistent—in the sense of Driscoll et al. [6], preserving previous versions after modification, not database persistence.
- (2) **Structural sharing:** Updates create new versions sharing structure with old.
- (3) **Object identity:** Distinguishes version identity (eq) from value equality (=).
- (4) **Generic functions:** Multiple dispatch over destructuring patterns.
- (5) **Meta-object protocol:** Introspection and extension via adapted MOP protocols.

### 2.2 Identity and Equality

Following Hickey’s epochal time model [12], FOL simplifies Common Lisp’s four-level equality hierarchy (eq/eql/equal/equalp) to two levels. eq tests *reference identity*—each functional update produces a new object, so (eq alice older-alice) returns false even when both represent “Alice.” FOL’s = tests *value equality* via generic dispatch: numbers compare with numeric coercion, persistent objects compare by structural comparison of storage maps (using fset=equal?), and strings compare by content—subsuming the roles of eql, equal, and equalp. Two independently constructed objects with identical slot values are = and may be used as map keys (hashing is based on storage structure, not reference identity).

### 2.3 Syntax and Readability

FOL adopts Clojure’s reader and definition syntax for literals and destructuring, as these are more concise and readable than Common Lisp’s. Persistent collections use literal syntax with Clojure semantics:

```
[1 2 3]           ; Vectors
{:name "Alice" :age 30} ; Maps
#{1 2 3 4}        ; Sets
```

```
(defclass <person> []
  [[name :type <string>]
   [age :type <number>]])

(defn summarize [{:keys [name age]]]
  (str name " is " age " years old"))
```

## 2.4 Persistent Object Protocol

By default, user-defined classes inherit from `<persistent-object>`:

```
(defclass <person> []
  [[name :type <string>]
   [age :type <number>]])

(def alice (make <person> :name "Alice" :age 30))
(def older-alice (assoc alice :age 31)) ; Returns new instance
(:age alice) ; => 30 (unchanged)
(:age older-alice) ; => 31

defclass supports standard CLOS options (:type, :initarg,
:initform, :reader). Slot values are stored in a persistent hash
map, enabling  $O(\log n)$  updates with structural sharing.
```

## 3 Architecture and Implementation

### 3.1 Collection Implementation

FOL wraps FSet [4] and Sycamore data structures for its collections:

- **Vectors** (`[1 2 3]`): 32-way branching trees supporting  $O(\log_{32} n)$  random access.
- **Maps** (`{:a 1 :b 2}`): Hash array mapped tries (HAMTs) with efficient key-value operations.
- **Sets** (`#{1 2 3}`): Weight-balanced binary trees for ordered iteration.
- **Lists**: Persistent cons cells with standard  $O(1)$  prepend and  $O(n)$  access.

FOL also provides thirteen additional collection classes from `<array-dict>` to `<dense-int-set>` to `<array>`. These collection classes have either been optimized to use the combinations of FSet and Sycamore data structures that provide the best performance at both small sizes ( $< 20$  elements) and at scale ( $> 1000$  elements) or have been hand-written in the case of collections not based on persistent structures (list, lazy-seq).

### 3.2 Runtime Representations

Most runtime representations in the transpiled code other than collections and user-defined objects utilize a Common Lisp base. Primitives (like `<bool>`, `<string>`, the numeric tower, and `<symbol>`) are unwrapped Common Lisp primitives; functions with their closures are simply Common Lisp functions; FOL derives its package structure and error handling model from Common Lisp. Garbage collection is also inherited from Common Lisp—old versions of user-defined objects are collected when unreferenced; structural sharing ensures only unshared portions are reclaimed.

### 3.3 MOP Protocol Adaptation

FOL’s persistent metaclass adapts rather than replaces the MOP. Table 1 summarizes the protocol disposition.

Introspection protocols (class-slots, class-precedence-list, slot-definition-\*) work unchanged because FOL’s class *structure* is standard CLOS—only slot *storage* is redirected. The validate-superclass

extension permits cross-metaclass inheritance (persistent classes may inherit from standard classes), but mixed hierarchies must respect the invariant that persistent slots use FSet or Sycamore storage while inherited standard slots remain mutable—finalization signals an error if this invariant cannot be maintained. This adaptation requires the MOP: persistence cannot be implemented as a library because transparent slot access demands slot-value-using-class redirection at the metaclass level.

## 4 Features

### 4.1 Clojure Features

FOL implements key Clojure abstractions:

- **Transducers** [13]: Composable transformation pipelines separating the “what” (map, filter, take) from the “how” (eager, lazy, parallel).
- **Lazy sequences**: On-demand computation enabling infinite sequences and efficient streaming.
- **Thread-safe atoms**: Mutable references with per-atom locking for concurrent updates.
- **Sequence functions**: Over 100 functions including map, filter, reduce, take, drop, partition, group-by, and frequencies.
- **Macros**: Clojure-style defmacro with syntax-quote (```), unquote (`~`), and splicing unquote (`~@`); macros receive unevaluated forms. Example: (defmacro unless [test & body] `(if (not ~test) (do ~@body))).
- **Atoms, Refs/Software Transactional Memory, and Agents**: Clojure’s concurrency primitives for managing mutable state in a thread-safe manner.
- **Reader syntax**: Clojure-style literals for vectors, maps, sets, keywords, metadata, etc.

### 4.2 Generic Function Integration

FOL’s generic functions dispatch by arity first, then by type predicates (`<vector>?`, `<dict>?`) and specificity within an arity group. Methods at different arities may coexist on the same generic:

```
(defgeneric process [x])
(defmethod process [(x <vector>)] (vec (map process x)))
(defmethod process [(x <dict>)] (update-vals x process))
(defmethod process [(x <number>)] (* x 2))
```

```
(process [1 {:a 2 :b 3} [[4]]]) ; => [2 {:a 4 :b 6} [[8]]]
```

```
;; Multi-clause methods with predicate dispatch
(defgeneric process ([a (b < 10)]) [a (b <number>) (c <string>)] [a
b]))
(defmethod process
  ([a (b (< 10))] (* 2 b))
  ([a (b <number>) (c <string>)] c)
  ([a b] (str a b)))
```

Figure 1 gives the formal grammar for FOL’s multi-pattern dispatch syntax.

**Predicate semantics:** The form (symbol (fn arg0 arg1 ...)) evaluates (fn symbol arg0 arg1 ...)—the bound value is inserted as the first argument. This works with functions, macros, or special forms.

**defn vs defgeneric/defmethod:** defn defines a standalone multi-clause function with predicate dispatch and specificity ordering; it

Table 1: MOP Protocol Adaptation

| Protocol                                                                                              | Status     | Rationale                                                  |
|-------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------|
| slot-value-using-class                                                                                | Redirected | Reads from Sycamore hash-map                               |
| (setf slot-value-using-class)                                                                         | Guarded    | Blocks post-init writes                                    |
| slot-boundp-using-class                                                                               | Redirected | Checks Sycamore hash-map membership                        |
| slot-makunbound-using-class                                                                           | Blocked    | Immutability invariant                                     |
| validate-superclass                                                                                   | Extended   | Cross-metaclass mixing                                     |
| initialize-instance                                                                                   | Extended   | Populates Sycamore hash-map from initargs                  |
| update-instance-for-redefined-class                                                                   | Omitted    | Instances are immutable; lazy migration instead (Sec. 5.4) |
| change-class                                                                                          | Omitted    | Violates immutability                                      |
| <i>Class computation, method combination, allocation: standard CLOS behavior inherited unchanged.</i> |            |                                                            |

```

defn ::= (defn name (single-clause | multi-clause))
defgeneric ::= (defgeneric name (single-pattern | multi-pattern))
defmethod ::= (defmethod name qualifier2 (single-clause | multi-clause))
fn ::= (fn (single-clause | multi-clause))
single-clause ::= pattern body+
multi-clause ::= (pattern body+)+
single-pattern ::= pattern
multi-pattern ::= pattern+
pattern ::= [param*] | [param* & rest-param]
param ::= symbol
           | (symbol type)
           | (symbol (fn-name arg*))
           | destructure
destructure ::= [param*]
              | { :keys [key-spec*] }
key-spec ::= symbol | (symbol type) | (symbol pred)
type ::= <type-name>
qualifier ::= :before | :after | :around

```

(simple binding)  
(type specialized)  
(predicate specialized)  
(nested destructuring)  
(sequential)  
(map)

Figure 1: FOL Multi-Pattern Dispatch Grammar (simplified)

does not create a generic function. `defgeneric` declares a generic function; `defmethod` adds methods to it. Only `defmethod` accepts `:before`/`:after`/`:around` qualifiers. Both `defn` and `defmethod` support multi-clause predicate dispatch.

**4.2.1 Method Combinations.** Multi-clause `defmethod` forms expand to N separate method definitions. The `:before`, `:after`, and `:around` qualifiers apply per-clause; `call-next-method` follows CLOS semantics.

### 4.3 Predicate Specializers

FOL extends pattern matching with predicate specializers. The syntax `(var (fn arg0 arg1 ...))` applies the predicate to the bound value at runtime.

**4.3.1 Predicate Examples.** Predicates work with any function—equality, comparison, type tests, or user-defined. In the first example below, all clauses are predicates (same category), so definition order determines dispatch; the second example mixes type predicates and a catch-all, where cross-category specificity applies:

```
;; Range checking with comparison predicates
```

```

(defn age-group
  ([ (age (< 13))] :child)
  ([ (age (< 20))] :teen)
  ([ (age (< 65))] :adult)
  ([age] :senior))

;; Type predicates and user-defined predicates
(defn process-value
  ([ (x (<number>?))] (* x 2))
  ([ (x (<string>?))] (str x x))
  ([ (x (<vector>?))] (vec (map process-value x)))
  ([x] x))

```

**4.3.2 Predicate Specificity.** FOL orders patterns by *constraint strength*—how narrowly a pattern constrains its accepted values. More specific patterns are tested first, ensuring predictable dispatch regardless of definition order:

This ordering is a deliberate heuristic chosen for predictability rather than full logical subsumption, which is undecidable in general [8]. The intuition is that a predicate like `(= 5)` accepts strictly fewer values than `<number>`, which accepts fewer than an untyped destructuring pattern. The boundary between type and predicate categories is conventional: type predicates like

Table 2: Predicate Specificity Levels

| Level | Pattern Type  | Constraint                | Example      |
|-------|---------------|---------------------------|--------------|
| 3     | Predicate     | Single value/narrow range | (n (= 5))    |
| 2     | Type          | All instances of a type   | (x <number>) |
| 1     | Destructuring | Structural shape only     | [a b c]      |
| 0     | Any           | Universal match           | x, _         |

(<number>?) are recognized syntactically and assigned level 2, while other predicates (including user-defined type tests) receive level 3. This means (x (<number>?)) and (x <number>) accept the same values but occupy different specificity levels if mixed: a clause using (<number>?) (level 3) will shadow one using <number> (level 2) for numeric arguments, even though both match the same set. In practice this is benign—programmers would use one form or the other, not both—but it is a pragmatic tradeoff we accept for uniform, predictable dispatch. Within a category, definition order is preserved, giving programmers explicit control over tie-breaking. For sequence patterns, specificity extends to element-level comparisons: [(x (< 0)) y] is more specific than [(x <number>) y] because its first element has higher specificity.

**4.3.3 Formal Specificity Ordering.** We formalize the specificity relation  $\prec$  using inference rules, shown in Figure 2. Let  $\mathcal{L}(p) \in \{0, 1, 2, 3\}$  be the specificity level of pattern  $p$ .

The operational dispatch order  $\prec_{\text{disp}}$  is a total order that resolves ties (where neither  $p_1 \prec p_2$  nor  $p_2 \prec p_1$ ) using definition order. Let  $\text{idx}(p)$  denote the definition index of the method clause containing  $p$ .

$$p_1 \prec_{\text{disp}} p_2 \iff p_1 \prec p_2 \vee (\neg(p_2 \prec p_1) \wedge \text{idx}(p_1) < \text{idx}(p_2))$$

Patterns of different arity are incomparable under  $\prec$ ; dispatch selects the arity group first, then applies  $\prec_{\text{disp}}$  within it. This lexicographic combination ensures deterministic dispatch: arity-first, then specificity, then definition-order.

**4.3.4 Dispatch Semantics and Type Checking.** Predicate specializers also work in anonymous fn forms but not in macro parameter lists. Dispatch is  $O(N)$  in the number of patterns at a given arity, with patterns pre-sorted by specificity. FOL’s type system is entirely dynamic—type specializers are checked at runtime. The compiler has not been extended to warn about unreachable clauses; a future compiler version could detect shadowing statically.

**4.3.5 Soundness Properties.** FOL’s dispatch maintains three invariants: (1) **exhaustive matching**—unmatched calls signal an error no-matching-clause rather than silently failing; (2) **deterministic dispatch**—the total order  $\prec_{\text{disp}}$  ensures exactly one clause matches; (3) **type predicate consistency**—type specializers use the same predicates as explicit checks. FOL does *not* provide static type soundness; errors are detected at runtime only.

## 5 Synergy in Practice

This section demonstrates patterns that are *more natural or more concise* in FOL than in either Clojure or standard CLOS alone, because they exploit the combination of persistent data, CLOS method combinations, and predicate dispatch. We focus on patterns that *require* multiple features working together—not merely features used in proximity but features whose interactions produce capabilities unavailable in either parent language. Full source for all examples is available in the repository: FOL code in fol-code/, Common Lisp equivalents in lisp-code/, and Clojure equivalents in clojure-code/.

### 5.1 Declarative Dispatch: Trade Compliance

We implement a trade compliance validator in FOL and plain Common Lisp (repository files compliance.fol, compliance.lisp). The FOL version uses predicate dispatch with automatic specificity ordering; the CL version uses a manual cond cascade with hand-ordered predicates. Table 3 compares the implementations (non-blank, non-comment lines).

```
;; Inherits from <persistent-object> for immutability
(defclass <trade> [<>]
  [[symbol :initarg :symbol :accessor trade-symbol]
   [amount :initarg :amount :accessor trade-amount]
   [price :initarg :price :accessor trade-price]
   [side :initarg :side :accessor trade-side]]) ; :buy or :sell

;; ...
(defn is-buy-side-trade?
  ([trade] (if (trade-side) (= :buy))) t)
  ([trade] nil))

(defn is-penny-stock?
  ([trade] (if (trade-price) (< 5.00))) t)
  ([trade] nil))

(defn penny-stock-buy? [trade]
  (and (is-buy-side-trade? trade) (is-penny-stock? trade)))

(defgeneric validate-trade [trade])
;; Syntax: (var predicate-fn)
(defmethod validate-trade
  ;; Rule 1: Block restricted symbols immediately
  ([trd (restricted-symbol?)])
  ;; ...
  ;; Rule 2: Flag high value trades for manual review
  ([trd (high-value?)])
  ;; ...
  ;; Rule 3: Warn on penny stock purchases
  ([trd (penny-stock-buy?)])
  { :status :warning
    :reason "High risk penny stock purchase"
    :trade trd })
  ;; Rule 4: Standard approval (Catch-all)
  ([trd]
   { :status :approved
     :id (gensym "TRD") }) )
```

Table 3 shows that savings in LOC are modest (17%) because this example exercises predicate dispatch but not persistence or method combinations—it demonstrates a single-feature advantage (predicate dispatch) rather than multi-feature synergy. The predicate category is *larger* in FOL because predicate-dispatch patterns



|                                       |                                         |                                                                           |
|---------------------------------------|-----------------------------------------|---------------------------------------------------------------------------|
| SPEC-LEVEL                            | SPEC-SEQ-HEAD                           | SPEC-SEQ-TAIL                                                             |
| $\mathcal{L}(p_1) > \mathcal{L}(p_2)$ | $p_1 \prec q_1 \quad n \geq 1$          | $\mathcal{L}(p_1) = \mathcal{L}(q_1) \quad [p_2 \dots] \prec [q_2 \dots]$ |
| $p_1 \prec p_2$                       | $[p_1 \dots p_n] \prec [q_1 \dots q_n]$ | $[p_1 \dots p_n] \prec [q_1 \dots q_n]$                                   |

Figure 2: FOL Predicate Specificity Inference Rules

Table 3: Code Comparison—Trade Compliance

| Category              | FOL       | CL        | Savings    |
|-----------------------|-----------|-----------|------------|
| Class definition      | 5         | 8         | 38%        |
| Predicates / logic    | 16        | 13        | −23%       |
| Dispatch / validation | 15        | 15        | 0%         |
| Module boilerplate    | 2         | 10        | 80%        |
| <b>Total</b>          | <b>38</b> | <b>46</b> | <b>17%</b> |

require explicit true/nil branches. The advantage here is *correctness*, not brevity: FOL’s specificity ordering guarantees that more-constrained rules fire first regardless of definition order. Reordering or adding rules cannot silently change which clause matches—a class of maintenance bug that plagues cond cascades in production.

## 5.2 Event Sourcing with Method Combinations

FOL enables event sourcing by combining `:around` methods with persistent event logs. The key mechanism is a single `:around` method that intercepts all command dispatches:

```
(defmethod apply-command :around [agg cmd]
  (bind [result (call-next-method)
        event {:command cmd :timestamp (now)}]
    (assoc result :events
      (conj (:events result) event))))
```

This `:around` method applies automatically to every `apply-command` invocation, including future methods added after deployment. The Clojure equivalent uses `defmulti/defmethod` but lacks `:around` methods, so event logging requires a manual wrapper that callers must remember to invoke. Table 4 compares the two (see repository for full source).

Table 4: Code Comparison—Event Sourcing

| Category        | FOL       | Clj       | Notes                           |
|-----------------|-----------|-----------|---------------------------------|
| Class / factory | 2         | 2         | Map literal                     |
| Event logging   | 4         | 4         | <code>:around</code> vs wrapper |
| Command methods | 7         | 5         | Pred. vs key dispatch           |
| Replay          | 2         | 2         | Identical                       |
| <b>Total</b>    | <b>15</b> | <b>14</b> |                                 |

Line counts are nearly identical, but the two implementations differ qualitatively. FOL’s `:around` method is *declarative*—it applies automatically to all invocations, including future methods; Clojure’s wrapper requires callers to remember to use `apply-command-logged`, and calling the multimethod directly silently skips logging. Persistence contributes a second guarantee: because `assoc` and `conj` produce new immutable values, the event log is append-only by

construction—no code path can retroactively alter a recorded event, which is the core invariant event sourcing requires. In mutable CLOS, both the ‘around’ interception and defensive copying of the event list would be needed to achieve the same guarantee.

## 5.3 Observable Slots via MOP and Predicate Dispatch

This example combines persistent functional updates with predicate dispatch to build a change-notification system (repository file `observable.fol`). When a slot is updated, the protocol produces both a new object (via persistent `assoc`) and a change event; predicate dispatch routes notifications by content—e.g., detecting reading spikes (>50 unit jump) or critical status transitions. The CL equivalent (`observable.lisp`) requires a manual copy function, a separate `defstruct` for change events, and a hand-coded cond cascade. Table 5 compares the implementations.

```
;; Domain class
(defclass <sensor> []
  [[name :initarg :name]
   [reading :initarg :reading :initform 0]
   [status :initarg :status :initform :normal]])

;; A change event is a persistent map produced by the update
protocol
(defn make-change [obj slot old-val new-val]
  {:object obj :slot slot :old old-val :new new-val})

;; Generic update that produces a change event alongside the new
object
(defn updated [obj slot new-val]
  (bind [old-val (slot slot obj)
        new-obj (assoc obj slot new-val)]
    {:object new-obj
     :change (make-change obj slot old-val new-val)}))

;; Predicate dispatch routes change notifications by content
(defgeneric on-change [change])

;; Spike detection: reading jumps by more than 50
(defmethod on-change
  ([c (-> :change (-> :slot (= :reading)))
   (-> :change (-> (fn [ch] (> (- (:new ch) (:old ch)) 50))))])
  {:alert :spike
   :message (str "Spike detected: "
                 (:old (:change c)) " -> " (:new (:change c))))})

;; Status transition to :critical
(defmethod on-change
  ([c (-> :change (-> :slot (= :status)))
   (-> :change (-> new (= :critical))))]
  {:alert :critical
   :message "Sensor entered critical state"})

;; Default: log the change
(defmethod on-change
```

```

581 ([c]
582   (:alert :info
583     :message (str "Slot " (:slot (:change c))
584                  " changed: " (:old (:change c))
585                  " -> " (:new (:change c))))))

```

Table reftab:observable shows a 40% reduction in lines of code. The CL version requires more boilerplate to achieve the same functionality, while FOL's combination of features enables a more concise and declarative implementation.

**Table 5: Code Comparison—Observable Slots**

| Category            | FOL       | CL        | Savings    |
|---------------------|-----------|-----------|------------|
| Class / boilerplate | 6         | 23        | 74%        |
| Update protocol     | 7         | 9         | 22%        |
| Change dispatch     | 18        | 20        | 10%        |
| <b>Total</b>        | <b>31</b> | <b>52</b> | <b>40%</b> |

The 40% reduction comes primarily from eliminating boilerplate: FOL needs no manual copy function (persistent assoc shares structure), no separate defstruct for change events (persistent maps suffice), and no package ceremony. This example best illustrates three-way synergy: persistence provides safe functional updates that produce change events without corrupting the original; predicate dispatch routes those events by content (e.g., reading spikes, critical transitions); and generic functions with defmethod enable new notification rules to be added without modifying the existing dispatch. None of these features alone suffices—Clojure has persistence but no method dispatch for extensible routing; CLOS has dispatch but requires manual deep copies to produce safe change events.

## 5.4 Lazy Schema Evolution

Standard CLOS handles class redefinition by lazily updating existing instances via `update-instance-for-redefined-class`: each instance is mutated in place the next time it is accessed. This protocol assumes mutable instances—an assumption FOL deliberately violates.

FOL's persistent metaclass resolves this tension through what amounts to *lazy schema evolution*. Class definitions remain mutable (redefining a class updates the class object normally), but existing instances are frozen snapshots whose %persistent-storage Sycamore hash-maps are never modified. The MOP's `slot-value-using-class` method provides graceful degradation: when accessing a slot that was added by a redefinition but is absent from an old instance's storage map, the method falls back to evaluating the slot's `:initform` if one exists, or signals `slot-unbound` otherwise. Slots removed by redefinition remain in the storage map but become inaccessible (no slot definition routes to them). Crucially, functional updates perform implicit migration of already existing objects. `set-pslot-value` calls `(class-of object)` to obtain the *current* (redefined) class, allocates a fresh instance of that class, and copies the old storage map forward. The resulting object is an instance of the new schema carrying the old data—new slots resolve via `initforms`, removed slots persist harmlessly in the map. Persistence makes this *safe* in a specific sense: old instances are never corrupted and remain

valid snapshots of the prior schema; in mutable CLOS, redefinition silently mutates every reachable instance. The tradeoff is that CLOS's `update-instance-for-redefined-class` provides an explicit migration hook where the programmer can transform slot values, while FOL's lazy approach silently returns `initform` defaults for new slots on old instances—a form of graceful degradation that could mask schema mismatches if the programmer is not aware. This lazy migration strategy is, to our knowledge, a novel adaptation of the MOP for immutable object systems—it trades explicit migration control for corruption safety and zero-downtime operation.

## 6 Evaluation

### 6.1 Comparison with Related Languages

**Table 6: Built-in Feature Comparison**

| Feature             | FOL | Clojure        | CL             |
|---------------------|-----|----------------|----------------|
| Persistent objects  | ✓   | ✓              | ×              |
| CLOS/MOP            | ✓   | ×              | ✓              |
| Method combinations | ✓   | ×              | ✓              |
| Predicate dispatch  | ✓   | ~ <sup>a</sup> | × <sup>b</sup> |
| Transducers         | ✓   | ✓              | ×              |
| Lazy seqs           | ✓   | ✓              | ×              |
| Macros              | ✓   | ✓              | ✓              |

<sup>a</sup> defmulti dispatches on an arbitrary fn but lacks per-parameter specializers and specificity ordering.

<sup>b</sup> EQL specializers only; optima/trivia provide pattern matching as libraries.

FOL uniquely combines Clojure's functional features with Common Lisp's object system. Persistent structures incur  $O(\log n)$  vs  $O(1)$  for mutable operations, but high branching factors (32–64) keep constant factors small.

### 6.2 Benchmark Methodology

All measurements were taken on SBCL 2.6.0 (Windows 11, AMD Ryzen 9 5900X, 64 GB RAM). Performance benchmarks report the mean of 100 runs (coefficient of variation <2%); memory benchmarks report single allocations via `sb-ext:dynamic-space-usage`. We compare against native Common Lisp rather than Clojure: SBCL compiles to native code, providing a strict upper bound on overhead, and since FOL's implementation is Common Lisp, this isolates abstraction cost. A Clojure comparison would conflate language design with JVM runtime differences (startup, JIT warmup, GC pauses).

### 6.3 Benchmarks

We evaluated FOL across four diverse benchmark categories to isolate the impact of its key features: predicate dispatch (Trade Compliance), observability patterns, event sourcing, and persistent state management (LSim).

**6.3.1 Feature-Specific Benchmarks.** Table 7 summarizes performance and code density across benchmarks focusing on specific FOL language features.

**Table 7: Feature Benchmark Summary**

| Benchmark          | Perf Ratio | FOL SLOC | CL SLOC | Ratio % |
|--------------------|------------|----------|---------|---------|
| Trade Compliance   | 2.16x      | 50       | 60      | 83%     |
| Sensor Observation | 5.88x      | 45       | 52      | 87%     |
| Event Sourcing     | 8.10x      | 28       | 45      | 62%     |

**Trade Compliance:** This benchmark exercises predicate dispatch without heavy persistent state updates. The 2.16x slowdown is largely due to Lisp-1 compatibility checks, while achieving a 17% reduction in SLOC.

**Sensor Observation:** Implements a monitoring loop with content-based notifications. The 5.88x overhead stems from producing multiple immutable event records per update. FOL reduces SLOC by 13% by eliminating manual copying.

**Event Sourcing:** This benchmark leverages ‘around’ methods and persistent logs. It shows the largest code density benefit (38% SLOC reduction) due to declarative command routing, albeit with an 8.1x performance premium for maintaining full audit trails.

**6.3.2 Logic Simulation (LSim).** LSim is a digital logic simulator that scales with circuit complexity. We evaluated three configurations: 8bit-100 (small), 32bit-300 (medium), and 8x32-900 (a 1024-gate pipeline). Table 8 shows that while FOL carries a constant-factor overhead, the performance ratio improves significantly as system complexity grows, narrowing from 11.3x to 3.2x. Beyond performance, FOL achieved a significant reduction in implementation effort, requiring 15–28% fewer lines of code across the core simulator and its test cases. This density is primarily due to the elimination of CLOS boilerplate and the use of concise collection literals.

**Table 8: LSim Scaling Results (Mean ms/run)**

| Configuration | CL (ms) | FOL (ms) | Ratio |
|---------------|---------|----------|-------|
| 8bit-100      | 1.0     | 11.3     | 11.3x |
| 32bit-300     | 8.4     | 58.1     | 6.9x  |
| 8x32-900      | 444.9   | 1425.5   | 3.2x  |

## 6.4 Threats to Validity

All measurements compare FOL’s transpiled code against SBCL’s native compiler, so reported ratios are comparable. As a single-author project, FOL has not yet undergone independent code review; the full source is publicly available to facilitate external evaluation.

## 7 Limitations and Future Work

FOL is transpiled to Common Lisp. No static analysis exists—type errors and unreachable clauses are detected at runtime; some type-dependent code may be resolved at run-time, degrading transpiled code’s performance. Memory overhead grows with object size (1.1x for 2 slots, 4x for 100 slots). Random updates incur 21x overhead versus  $O(1)$  mutation. Class redefinition does not migrate existing instances (Section 5.4); old objects retain their original slot data, with new slots resolved lazily via `initforms`.

Planned extensions include **abstract and sealed classes**, **parallel collections** following Scala’s model, and **channel-based concurrency** via CSP-style go blocks.

## 8 Related Work

**Clojure** [11] pioneered persistent data structures in Lisp. Clojure’s *protocols* provide type-based polymorphism (single dispatch on the first argument’s type) as a lightweight alternative to CLOS generic functions. FOL adopts Clojure’s sequence abstraction and transducers but uses CLOS generic functions with multiple dispatch and method combinations—gaining `:before/after/around` qualifiers and MOP introspection at the cost of Clojure’s protocol performance optimizations.

**Common Lisp’s** CLOS [3] and MOP [14] provide FOL’s object system foundation, extended with persistent slots. **Dylan** [1] was an early implementation of an object-oriented Scheme. In FOL, it influenced naming conventions (`<type>` notation).

**Persistent data structures** build on Okasaki [15]; FSet [4] provides production-quality collections for Common Lisp. FOL’s functional update pattern is related to **lenses** in the Haskell tradition; FOL does not yet provide a lens abstraction. **Clojure’s** `defrecord`, Scala case classes, and Kotlin data classes all provide value-oriented objects, but none integrate with a MOP—FOL is unique in combining persistent storage with metaclass-level extensibility.

### 8.1 Predicate Dispatch and Pattern Matching

Ernst et al. [8] introduced predicate dispatch with logical implication for specificity. Chambers and Chen [5] demonstrated efficient implementation strategies using dispatch DAGs, achieving performance competitive with hand-written type-case dispatch. **Filtered Dispatch** [16] extended CLOS with predicate-based method selection using explicit priorities. FOL’s category-based specificity with first-match tie-breaking provides a middle ground between logical implication (expressive but undecidable) and explicit priorities (simple but error-prone). **Scala’s** `extractors` [7] and **Racket’s** `match expanders` provide extensible pattern matching; **OCaml’s** `polymorphic variants` [10] offer extensible sum types. FOL differs: predicates are arbitrary functions rather than structural extractors, and specificity is determined by category rather than type inclusion.

### 8.2 Multiple Dispatch and Typed Functional Lisps

**Julia** [2] uses multiple dispatch but is purely type-based, lacking predicate specializers, method combinations, and MOP introspection. **Coaltion** [17] adds Hindley-Milner types to CL—complementary to FOL (static safety vs. persistence). **Typed Racket** [20] adds gradual typing; FOL’s type specializers serve as dispatch mechanisms rather than correctness guarantees. **Shen** [19] and **Carp** [18] offer pattern matching and static typing respectively, but lack multiple dispatch with method combinations. **Racket’s** `class system` [9] provides method combinations but separates classes from match; FOL unifies pattern matching and method dispatch. **Elixir** combines pattern matching with protocols on the BEAM VM but lacks multiple dispatch and method combinations.

## 9 Conclusion

FOL demonstrates that persistent objects and CLOS are synergistic rather than competing paradigms. Their combination yields versioned objects via persistent snapshots, declarative event sourcing via method combinations over persistent logs, and observable slots via predicate dispatch over change events—patterns that require multiple features working together and are more natural than in either parent language alone. Lazy schema evolution (Section 5.4) further illustrates how immutability transforms a traditionally problematic MOP protocol into a corruption-safe migration strategy, albeit at the cost of CLOS’s explicit migration hooks. Benchmarks confirm that interpretation, not persistence, dominates current overhead (300–500x vs 1–3x). A preliminary transpiler to Common Lisp with a single optimization—fixed-arity dispatch—reaches parity with hand-written code (0.93x) on the compliance benchmark, eliminating the dispatch overhead entirely. This demonstrates that compilation, not language design, is the sole performance bottleneck, and that a complete compiler can retain immutability’s correctness and concurrency benefits at native speed. The complete implementation—385 functions, 6,333 tests, and ~18,000 lines of bootstrap code—is available at: <https://github.com/frankadrian314159/fol>

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